On degenerations of $\mathcal{D}^\times$-shtukas over ramified legs (Joint with Wansu Kim and Junyeong Park)

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Shimura varieties

Modular curve

A modular curve (as a complex analytic stack)

$$Y(1)^{\mathsf{an}} = [\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}].$$

There is an equivalence of groupoids

$$Y(1)^{\mathsf{an}} \leftrightarrow \{\text{elliptic curves over } \mathbb{C}\}.$$

This leads to a smooth separated DM stack $\mathcal{Y}(1)$ of finite type over $\mathbb{Z}$ with

$$\mathcal{Y}(1)(S) = \{\text{elliptic curve over } S\} \quad \text{for } S \in \mathsf{Sch}_{\mathbb{Z}},$$

so $\mathcal{Y}(1)(\mathbb{C}) = Y(1)^{\mathsf{an}}$.

Shimura curves

D: quaternion division $\mathbb{Q}$-algebra D with fixed $D \otimes_{\mathbb{Q}} \mathbb{R} \simeq M_2(\mathbb{R})$.
$\mathcal{D} \subseteq D$: maximal $\mathbb{Z}$-order

A Shimura curve

$$X^D(1)^{\mathsf{an}} = [\mathcal{D}^{\times}_{\mathsf{Nrd}=1} \backslash \mathbb{H}].$$

There is an equivalence of gruopoids

$$X^D(1)^{\mathsf{an}} \leftrightarrow \{(A, \iota)\}$$

where A is an abelian surface over $\mathbb{C}$, and $\iota : \mathcal{D} \hookrightarrow \mathrm{End}_{\mathbb{C}}(A)$ is an injective ring map ('QM').

Shimura curves

This leads to a separated DM stack $\mathcal{X}^D(1)$ of finite type over $\mathbb{Z}$ with

$$\mathcal{X}^D(1)(S) = \{\text{abelian surfaces over } S \text{ with QM}\} \quad \text{for } S \in \mathrm{Sch}_{\mathbb{Z}},$$

so $\mathcal{X}^D(1)(\mathbb{C}) = X^D(1)^{\mathrm{an}}$.

However, it is smooth only over $\mathbb{Z}[1/\operatorname{disc}(D)]$ (Čerednik–Drinfeld).

Shimura varieties

For a reductive group G over $\mathbb{Q}$ (with auxiliary data X and level K), there is a complex manifold

$$\mathrm{Sh}_K(G,X)^{\mathrm{an}} \overset{\text{roughly}}{\approx} \coprod_{\text{finite}} \frac{\text{Hermitian symmetric domain}}{\text{arithmetic subgroup of } G(\mathbb{Q})}$$

called a Shimura variety for G.
This complex manifold $\mathrm{Sh}_K(G,X)^{\mathrm{an}}$ has a 'canonical' smooth scheme

$$\mathrm{Sh}_K(G,X)$$

over a number field E with

$$\mathrm{Sh}_K(G,X)(\mathbb{C}) = \mathrm{Sh}_K(G,X)^{\mathrm{an}}.$$

Warning: $\mathrm{Sh}_K(G,X)$ may not have a nice and useful moduli description.

Shimura varieties

Nonetheless, almost every Shimura variety admits a 'canonical' scheme

$$\mathscr{S}_K(G, X)$$

over $\mathcal{O}_{E,(v)}$ (for place $v \in \mathrm{Spec}(\mathcal{O}_E)$ over a prime p) with

$$\mathscr{S}_K(G, X)_E = \mathrm{Sh}_K(G, X)$$

with a suitable condition on level K at p (analogous to $\Gamma_0(p)$ condition).

Theory of canonical integral models

- Kottwitz ('90's), Vasiu and Kisin ('00's),
- W. Kim–K. Madapusi Pera, Kisin–Pappas ('10's),
- Kisin–R. Zhou, Pappas–Rapoport,
  P. Daniels–P. v. Hoften–D. Kim–M. Zhang,
  P. Daniels–A. Youcis ('21-24)

Shimura varieties

1. $\mathcal{Y}(1)$ is a canonical integral model $\mathcal{S}_K(G, X)$ for $G = \mathrm{GL}_{2,\mathbb{Q}}$.
2. $\mathcal{X}^D(1)$ is a canonical integral model $\mathcal{S}_K(G, X)$ for $G = D^\times$.

Fact (Baily–Borel–Satake)

$\mathrm{Sh}_K(G, X)$ is proper over E if and only if G^{ad} is $\mathbb{Q}$-anisotropic.

Conjecture (Morita principle)

$\mathrm{Sh}_K(G, X)$ is proper over E if and only if $\mathscr{S}_K(G, X)$ is proper over $\mathcal{O}_{E,(v)}$.

This conjecture is verified in many cases (including the prev. cases).

1. $\mathrm{SL}_{2,\mathbb{Q}}$ is not $\mathbb{Q}$-anisotropic and $\mathcal{Y}(1) \to \mathrm{Spec}(\mathbb{Z})$ is not proper.
2. $\mathcal{D}^\times_{\mathrm{Nrd}=1}$ is $\mathbb{Q}$-anisotropic and $\mathcal{X}^D(1) \to \mathrm{Spec}(\mathbb{Z})$ is proper.

Moduli Spaces of Shtukas

Function field analogy

Number field	Function field
$\mathbb{Q}$	$\mathbb{F}_q(t)$
$\mathrm{Spec}(\mathcal{O}_E) \cup \{\text{arch. places}\}$	$X = \mathbb{P}^1_{\mathbb{F}_q}$
quat. div. alg. $D/\mathbb{Q}$	quat. div. alg. $D/\mathbb{F}_q(X)$
maximal order $\mathcal{D}$ in D	maximal order $\mathcal{D}$ over X

From now on, we adopt notations on the right.

Question

Number field	Function field
ab. surf. (A, ι) over with QM by $\mathcal{D}$	??

$\mathcal{D}^\times$-shtukas

- $S \in \mathrm{Sch}_{\mathbb{F}_q}$,
- $\mathcal{E}$ is locally free $\mathcal{D} \boxtimes \mathcal{O}_S$-module (on $X \times S$) with rank 1,
- $x_1, x_2 \in X(S)$ (called legs of shtuka),
- $\phi : (\mathrm{id}_X \times \mathrm{Frob}_S)^* \mathcal{E} \dashrightarrow \mathcal{E}$: isomorphism away from graphs Γ_{x_1}, Γ_{x_2}.

A $\mathcal{D}^\times$-shtuka over S with 2 legs is such tuple $(\mathcal{E}, x_1, x_2, \phi)$.

Bound

- theory of elementary divisor at unramified legs
- BD Schubert variety at ramified legs Arasteh Rad–Hartl, Bieker ('14–23)

$\mathcal{D}^\times$-shtukas

Let $Z \subsetneq X$ be the ramified locus of $\mathcal{D}$.

Number field	**Function field**
arch. place ∞	(fixed) $x_1 \in X$ with $\mathrm{inv}_{x_1}(D) = 0$
$\mathrm{Spec}(\mathbb{Z})$	$X - \{x_1\}$
$\mathrm{Spec}(\mathbb{Z}[1/\operatorname{disc}(D)])$	$X - Z - \{x_1\}$
ab. surf. (A, ι) over $S/\mathbb{Z}$ with QM by $\mathcal{D}$	$\mathcal{D}^\times$-shtuka over $S/\mathbb{F}_q$ with legs x_1 and $x_2 \in X - Z - \{x_1\}$ bounded by $(0, -1)$ and $(1, 0)$

$\mathcal{D}^\times$-shtukas

The moduli stack of $\mathcal{D}^\times$-shtukas bounded by μ is

$$\begin{aligned} \mathrm{Sht}^{\leqslant\mu}_{\mathcal{D}^\times} : \mathrm{Sch}^{\mathrm{op}}_{\mathbb{F}_q} &\to \mathrm{Grpd} \\ S &\mapsto \{(\mathcal{E}, x_1, x_2, \phi) \mid \phi \text{ bounded by } (\mu^{(1)}, \mu^{(2)})\}. \end{aligned}$$

$\mathrm{Sht}^{\leqslant\mu}_{\mathcal{D}^\times}$ is a separated DM stack over X^2

$$\begin{aligned} \mathrm{Sht}^{\leqslant\mu}_{\mathcal{D}^\times} &\to X^2 \\ (\mathcal{E}, x_1, x_2, \phi) &\mapsto (x_1, x_2), \end{aligned}$$

It is a countable disjoint union of finite type substacks.

We will implicitly consider the connected components of $\mathrm{Sht}^{\leqslant\mu}_{\mathcal{D}^\times}$.

$\mathcal{D}^\times$-shtukas

For $X' = \{x_1\} \times (X - \{x_1\})$,

$$\mathcal{X}^D(1) \to \mathrm{Spec}(\mathbb{Z}) \overset{analogous}{\approx} \mathrm{Sht}^{\leqslant\mu}_{\mathcal{D}^\times}|_{X'} \to X'$$

For $X'' = \{x_1\} \times (X - Z - \{x_1\})$.

$$\mathcal{X}^D(1) \to \mathrm{Spec}(\mathbb{Z}[1/\operatorname{disc}(D)]) \overset{analogous}{\approx} \mathrm{Sht}^{\leqslant\mu}_{\mathcal{D}^\times}|_{X''} \to X''$$

Properness of $\mathrm{Sht}^{\leqslant\mu}_{\mathcal{D}^\times}$

Theorem (Lau)

The morphism

$$\mathrm{Sht}^{\leqslant\mu}_{\mathcal{D}^\times}|_{(X-Z)^2} \to (X-Z)^2$$

is proper if and only if

$$\mu_1^{(1)} + \mu_1^{(2)} < \sum_{x\in Z}[\mathrm{inv}_x(D)]_{\mathbb{Q}}.$$

Here, $[a]_{\mathbb{Q}}$ *for* $a \in \mathbb{Q}/\mathbb{Z}$ *means the unique rational number with* $0 \leq [a]_{\mathbb{Q}} < 1$ *and* $[a]_{\mathbb{Q}} \equiv a \mod \mathbb{Z}$.

For the bounds $(0,-1)$ and $(1,0)$, the inequality holds if and only if D is ramified at ≥ 4 places.

Properness of $\mathrm{Sht}^{\leqslant\mu}_{\mathcal{D}^\times}$

Theorem (C., Wansu Kim, Junyeong Park)

Suppose the strong non-emptiness condition on 'supersingular locus' at ramified legs. Then the morphism

$$\mathrm{Sht}^{\leqslant\mu}_{\mathcal{D}^\times} \to X^2$$

is proper if and only if

$$\mu_1^{(1)} + \mu_1^{(2)} < \left\lfloor \frac{\deg^-(\mu^{(1)}) + \deg^-(\mu^{(2)})}{2} \right\rfloor + \sum_{x \in Z} [\mathrm{inv}_x(D)]_{\mathbb{Q}},$$

$$\textit{where} \qquad \deg^-(\mu^{(i)}) = \begin{cases} \mu_1^{(i)} + \mu_2^{(i)} & \textit{if it's } \leq 0 \\ 0 & \textit{otherwise.} \end{cases}$$

For the bounds $(0, -1)$ and $(1, 0)$, the inequality holds if and only if D is ramified at ≥ 6 places.

Properness of $\mathrm{Sht}_{\mathcal{D}^\times}^{\leqslant\mu}$

Remark

- The function field analogue of the Morita principle fails.
- The additional red term is strictly negative if $\mathrm{Sht}_{\mathcal{D}^\times}^{\leqslant\mu} \neq \varnothing$.

Strong non-emptiness of supersingular locus

Any shtuka can be specialized to a shtuka that is 'supersingular' at each ramified place.

Status

- The expected optimal inequality $\Rightarrow$ Properness: Conditional. Working on removing the hypothesis.
- A weaker inequality $\Rightarrow$ Properness: Unconditional
- Failure of the expected optimal inequality $\Rightarrow$ Non-properness: Unconditional.

Generalization

- For $\mathcal{D}^\times$-shtukas with any number of legs, a very similar statement is proved.
- For a general central division F-algebra D, a very similar statement is proved. (It is maybe harder to remove the strong non-emptiness condition.)
- Expect the result to be generalized to unitary case.

What's new compared to Lau?

- Definition of moduli stack of shtukas over X^2: Quite involved, requires geometry of Beilinson–Drinfeld Schubert varieties.
- Lau's proof can be adopted with some hard work for $\mathcal{D}^\times$-shtukas 'supersingular' at each ramified leg, e.g. understanding of Vinberg monoids (i.e. complete homomorphisms).

Application and Perspective

Application

The Mantovan formula for function fields allowing ramified legs.

Question

Can this result be applicable to intersection theory of special cycles?

Thanks for your attention!